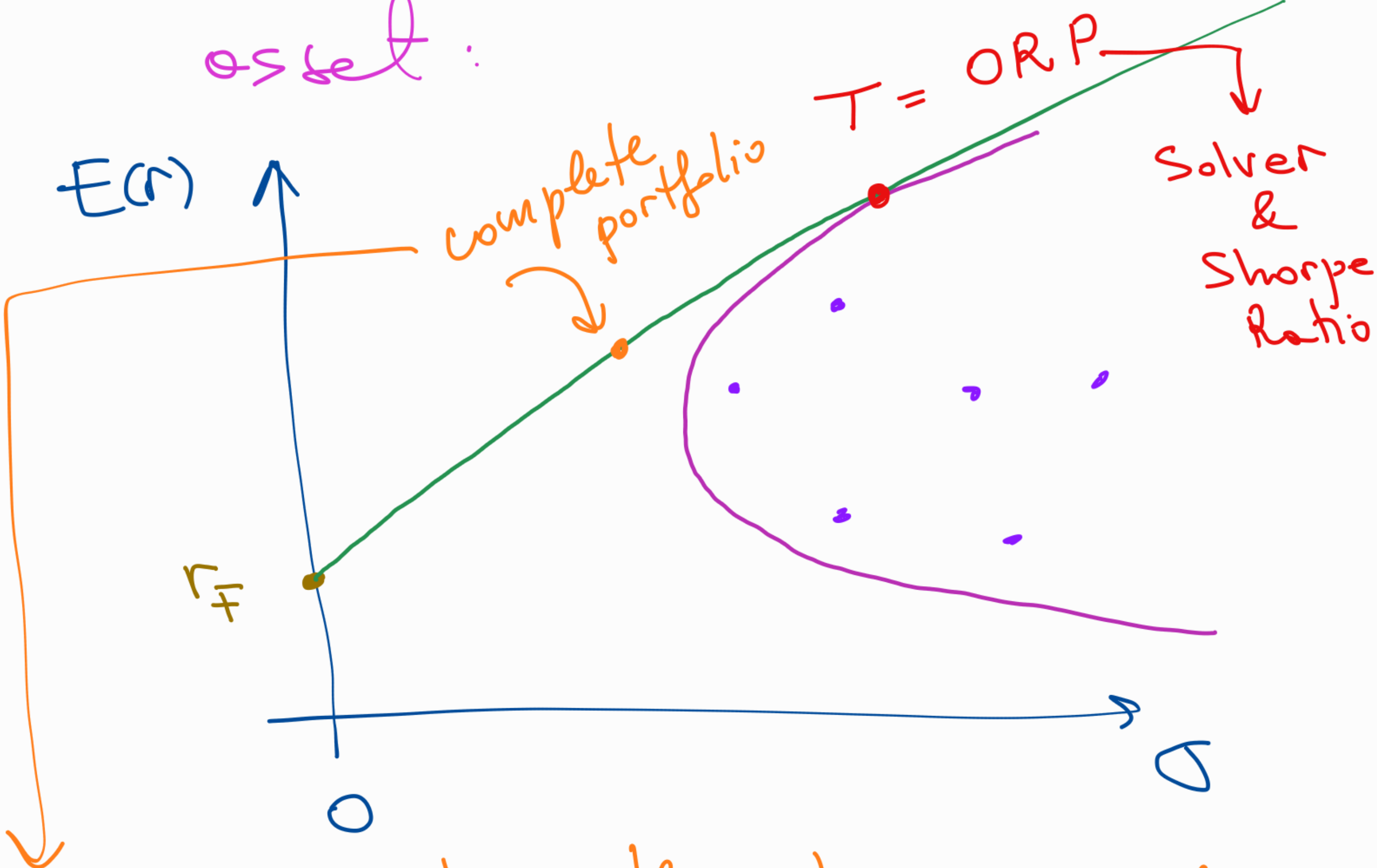
Capital Asset Pricing Model

(CAPM)

① Optimal Portfolio Allocation

Investors take prices/returns as given, and decide how much to buy of each asset:



for an investor with risk aversion, A:

$$y = \frac{E(r_T) - r_f}{A \times \sigma^2(r_T)}$$

(2) All investors place their orders to buy/sell assets $\Rightarrow$ prices adjust! How?

Suppose investor k has z_k in capital

Investor k will allocate z_k across the assets according to the weights in step (1)

Suppose the weight of asset i in ORP is w_i

$$z_k \times w_i \times \frac{E(r_T) - r_F}{A_k \times \sigma^2(r_T)}$$

Aggregate demand for asset i:

$$\sum_k z_k \times w_i \times \frac{E(r_T) - r_F}{A_k \times \sigma^2(r_T)}$$

$$D_i = w_i \times \frac{E(r_T) - r_F}{A_K \times \sigma^2(r_T)} \times \sum_K \frac{z_K}{A_K}$$

aggregate demand for asset i

Aggregate supply of asset i =

$$N_i \times p_i$$

shares outstanding

price per share

Demand = Supply

$$w_i \times \frac{E(r_T) - r_F}{A_K \times \sigma^2(r_T)} \times \sum_K \frac{z_K}{A_K} = N_i \times p_i$$

Sum across all assets i:

$$\left(\sum_i w_i\right) \times \frac{E(r_T) - r_F}{A_K \times \sigma^2(r_T)} \times \sum_K \frac{z_K}{A_K} =$$

$$\parallel$$

$$= \sum_i N_i \times p_i$$

$$\boxed{\frac{E(r_T) - r_F}{A_K \times \sigma^2(r_T)} \times \sum_K \frac{z_K}{A_K}} = \sum_i N_i \times p_i$$

$$\Downarrow$$

$$w_i \times \boxed{\frac{E(r_T) - r_F}{A_K \times \sigma^2(r_T)} \times \sum_K \frac{z_K}{A_K}} = N_i \times p_i$$

$$w_i \times \sum_i N_i \times p_i = N_i \times p_i$$

$$\boxed{w_i = \frac{N_i \times p_i}{\sum_i N_i \times p_i}}$$

Weights of ORP become the weights of S&P 500 (value-weighted).